

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
  <title>Class Timetable</title>
  <style>
    table {
      border-collapse: collapse;
      width: 80%;
      margin: 20px auto;
    }
    th, td {
      border: 1px solid black;
      text-align: center;
      padding: 8px;
    }
    th {
      background-color: #f2f2f2;
    }
  </style>
</head>
<body>
  <h1 style="text-align: center;">Class Timetable</h1>
  <table>
    <tr>
      <th>Day</th>
      <th>9:00-10:00</th>
      <th>10:00-11:00</th>
      <th>11:00-12:00</th>
      <th>1:00-2:00</th>
      <th>2:00-3:00</th>
    </tr>
    <tr>
      <td>Monday</td>
      <td>Math</td>
      <td>English</td>
      <td>Physics</td>
      <td>Chemistry</td>
      <td>Computer Science</td>
    </tr>
    <tr>
      <td>Tuesday</td>
      <td>History</td>
```

```
<td>Math</td>
<td>English</td>
<td>Biology</td>
<td>Sports</td>
</tr>
<tr>
<td>Wednesday</td>
<td>Physics</td>
<td>Chemistry</td>
<td>Computer Science</td>
<td>Math</td>
<td>English</td>
</tr>
<tr>
<td>Thursday</td>
<td>Biology</td>
<td>Math</td>
<td>History</td>
<td>Physics</td>
<td>Sports</td>
</tr>
<tr>
<td>Friday</td>
<td>English</td>
<td>Chemistry</td>
<td>Math</td>
<td>History</td>
<td>Biology</td>
</tr>
</table>
</body>
</html>
```

PROGRAM

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Three Frames Layout</title>
</head>
<frameset cols="20%, 60%, 20%">
  <frame src="contents.html" name="contents">
  <frame src="body.html" name="body">
  <frame src="remarks.html" name="remarks">
</frameset>
</html>
```

contents.html (Left Frame)

```
<html>
<head>
  <title>Contents</title>
</head>
<body>
  <h3>Contents</h3>
  <ul>
    <li>Introduction</li>
    <li>Services</li>
    <li>Contact</li>
  </ul>
</body>
</html>
```

body.html (Center Frame)

```
<!DOCTYPE html>
<html>
<head>
  <title>Body</title>
</head>
<body>
  <h2>Welcome to Our Website</h2>
  <p>This is the main content of the page.</p>
</body>
</html>
```

remarks.html (Right Frame)

```
<!DOCTYPE html>
<html>
<head>
  <title>Remarks</title>
</head>
<body>
  <h3>Remarks</h3>
  <p>Your feedback is valuable to us.</p>
</body>
</html>
```

PROGRAM:

```
<!DOCTYPE html>
<html >
<head>
  <title>CSS Demonstration</title>
  <!-- Linking External CSS -->
  <link rel="stylesheet" href="styles.css">
  <!-- Internal CSS -->
  <style>
    h1 {
      color: green;
      text-align: center;
    }
    .internal {
      font-size: 18px;
      color: blue;
    }
  </style>
</head>
<body>
  <!-- Inline CSS -->
  <h1 style="font-family: Arial, sans-serif;">Welcome to CSS Demonstration</h1>
  <p class="internal">This paragraph is styled using internal CSS</p>
  <p class="external">This paragraph is styled using external CSS</p>
  <p style="color: red; font-weight: bold;">This paragraph is styled using inline CSS.</p>
</body>
</html>
```

Style.css

```
.external {
  font-size: 20px;
  color: purple;
  font-style: italic;
}
```

PROGRAM:

```
<!DOCTYPE html>
<html >
<head>
  <title>Embed Audio and Video</title>
</head>
<body>
  <h1>Embed Audio and Video in HTML</h1>
  <h2>Audio Example</h2>
  <audio controls>
    <source src="audio.mp3" type="audio/mpeg">
    Your browser does not support the audio element.
  </audio>

  <h2>Video Example</h2>
  <video controls width="480">
    <source src="video.mp4" type="video/mp4">
    Your browser does not support the video element.
  </video>
</body>
</html>
```

PROGRAM:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Employee Registration Form</title>
<style>
  body {
    font-family: Arial, sans-serif;
    background-color: #f4f4f4;
  }
  .container {
    width: 50%;
    margin: 50px auto;
    padding: 20px;
    background: #fff;
  }
  h1 {
    text-align: center;
    color: #333;
  }
  form
  {
    display: flex;
    flex-direction: column;
  }
  label {
    margin: 10px 0 5px;
  }
  input, select, textarea
  {
    padding: 10px;
    margin-bottom: 20px;
    border: 1px solid #ccc;
    border-radius: 5px;
  }
  input[type="submit"], input[type="reset"] {
    background-color: #28a745;
    color: #fff;
  }
  input[type="reset"] {
    background-color: #dc3545;
  }
  input[type="submit"]:hover {
    background-color: #218838;
```

```

    }
</style>
<script>
function validateform()
{
    var name = document.forms["regform"]["Name"].value;
    var email = document.forms["regform"]["Email"].value;
    var phone = document.forms["regform"]["Telephone"].value;
    var empid = document.forms["regform"]["Empid"].value;
    var password = document.forms["regform"]["Password"].value;
    var address = document.forms["regform"]["Address"].value;
    if (name == "") {
        alert("Please enter your name.");
        return false;
    }
    if (empid == "" || isNaN(empid))
    {
        alert("Please enter a valid Employee ID.");
        return false;
    }
    if (email == "")
    {
        alert("Please enter a valid e-mail address.");
        return false;
    }
    if (phone == "")
    {
        alert("Please enter your phone number.");
        return false;
    } else if (phone.length !== 10 || isNaN(phone))
    {
        alert("Please enter a valid 10-digit phone number.");
        return false;
    }
    if (password == "")
    {
        alert("Please enter your password.");
        return false;
    }
    else if (password.length < 8)
    {
        alert("Password must be at least 8 characters long.");
        return false;
    }
    if (address == "") {
        alert("Please enter your address.");
    }
}

```

```

        return false;
    }
    alert("Details sent successfully!");
    return true;
}
</script>
</head>
<body>
<div class="container">
    <h1>Employee Registration Form</h1>
    <form name=" regform" onsubmit="return validateform()" method="post">
        <label for="Empid">Employee ID:</label>
        <input type="text" name=" Empid " id=" Empid ">
        <label for="Name">Name:</label>
        <input type="text" name="Name" id="Name">
        <label for="Address">Address:</label>
        <input type="text" name="Address" id="Address">
        <label for="Email">E-mail Address:</label>
        <input type="text" name="Email" id="Email">
        <label for="Password">Password:</label>
        <input type="password" name="Password" id="Password">
        <label for="Telephone">Phone Number:</label>
        <input type="text" name="Telephone" id="Telephone">
        <label for="Subject">SELECT YOUR COURSE</label>
        <select name="Subject" id="Subject">
            <option value="">Select</option>
            <option value="BTECH">BTECH</option>
            <option value="BCA">BCA</option>
            <option value="B.COM">B.COM</option>
        </select>
        <label for="Comment">Comments:</label>
        <textarea name="Comment" id="Comment" cols="30" rows="5"></textarea>
        <input type="submit" value="Send" name="Submit">
        <input type="reset" value="Reset" name="Reset">
    </form>
</div>
</body>
</html>

```


PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Calculator</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 50px;
    }
    input, button {
      margin: 10px;
      padding: 10px;
      font-size: 16px;
    }
    #result {
      font-size: 18px;
      font-weight: bold;
    }
  </style>
</head>
<body>
  <h1>Simple Calculator</h1>
  <input type="number" id="num1" placeholder="Enter first number">
  <input type="number" id="num2" placeholder="Enter second number">
  <br>
  <button onclick="calculate('add')">Add</button>
  <button onclick="calculate('subtract')">Subtract</button>
  <button onclick="calculate('multiply')">Multiply</button>
  <button onclick="calculate('divide')">Divide</button>
  <button onclick="resetCalculator()">Reset</button>
  <p id="result">Result: </p>

  <script>
    function calculate(operation) {
      const num1 = parseFloat(document.getElementById('num1').value);
      const num2 = parseFloat(document.getElementById('num2').value);
      let result;
      if (isNaN(num1) || isNaN(num2)) {
        document.getElementById('result').innerText = "Please enter valid numbers.";
        return;
      }
      switch (operation) {
        case 'add':
          result = num1 + num2;
```

```
        break;
    case 'subtract':
        result = num1 - num2;
        break;
    case 'multiply':
        result = num1 * num2;
        break;
    case 'divide':
        if (num2 === 0) {
            document.getElementById('result').innerText = "Error: Division by zero.";
            return;
        }
        result = num1 / num2;
        break;
    default:
        result = "Invalid operation.";
    }
    document.getElementById('result').innerText = "Result: " + result;
}
function resetCalculator() {
    document.getElementById('num1').value = "";
    document.getElementById('num2').value = "";
    document.getElementById('result').innerText = "Result: ";
}
</script>
</body>
</html>
```

PROGRAM:

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Array Sorting</title>
</head>
<body>
  <h1>Simple Array Sorting</h1>
  <label>Enter numbers (comma separated):</label>
  <input type="text" id="numbers" placeholder="e.g., 4,2,7,1,5">
  <button onclick="performSort()">Sort Array</button>
  <h3>Sorted Array:</h3>
  <p id="output"></p>
  <script>
    function sortArray(arr) {
      for (let i = 0; i < arr.length - 1; i++) {
        for (let j = 0; j < arr.length - 1 - i; j++) {
          if (arr[j] > arr[j + 1]) {
            let temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
          }
        }
      }
      return arr;
    }
    function performSort() {
      let input = document.getElementById("numbers").value;
      let array = input.split(',').map(Number);
      let sortedArray = sortArray(array);
      document.getElementById("output").innerText = sortedArray.join(", ");
    }
  </script>
</body>
</html>
```

PROGRAM:

app.js

```
var app = angular.module('myApp', []);
app.controller('myCtrl', function($scope) {
    $scope.filteredDept = [];

    $scope.employees = [
        {'id': 1, 'dept': 'IT', 'sal': 5000, 'name': 'Ramesh'},
        {'id': 2, 'dept': 'IT', 'sal': 4000, 'name': 'Girish'},
        {'id': 3, 'dept': 'Management', 'sal': 10000, 'name': 'Kartik'},
        {'id': 4, 'dept': 'Marketing', 'sal': 8000, 'name': 'Shenoy'},
        {'id': 5, 'dept': 'Marketing', 'sal': 7000, 'name': 'Darvik'},
        {'id': 6, 'dept': 'Management', 'sal': 17000, 'name': 'Ravi Kumar'}
    ];

    $scope.empFiltered = $scope.employees;

    $scope.departments = [
        {'id': 1, 'name': 'IT', 'selected': false},
        {'id': 2, 'name': 'Management', 'selected': false},
        {'id': 3, 'name': 'Marketing', 'selected': false}
    ];

    $scope.sortEmp = function(dept_id) {
        $scope.departments.filter(function(e) {
            if (e.id === dept_id) {
                var idx = $scope.filteredDept.indexOf(e.name, 0);
                if (idx === -1) {
                    $scope.filteredDept.push(e.name);
                } else {
                    $scope.filteredDept.splice(idx, 1);
                }
            }
            $scope.filterEmpList();
        });
    };

    $scope.filterEmpList = function() {
        $scope.filteredData = [];
        if ($scope.filteredDept.length !== 0) {
            $scope.empFiltered = [];
            angular.forEach($scope.filteredDept, function(val) {
                $scope.filteredData = $scope.employees.filter(function(e) {
                    if (e.dept === val) {
                        return e;
                    }
                });
            });
            angular.forEach($scope.filteredData, function(v) {
```

```

        $scope.empFiltered.push(v);
    });
});
} else {
    $scope.empFiltered = $scope.employees;
}
};

$scope.hikeSal = function() {
    $scope.empFiltered.map(function(e) {
        if (e.hike === undefined) {
            e.hike = 0;
        }
        e.hike = parseInt(e.hike) + $scope.hike;
        e.hikeAmt = e.sal * ($scope.hike * 0.01);
        e.sal += e.sal * ($scope.hike * 0.01);
    });
};
});
});

```

Index.html

```

<!DOCTYPE html>
<html lang="en" ng-app="myApp">
<head>
    <meta charset="UTF-8">
    <title>Payroll Management System</title>
    <script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.6.4/angular.min.js"></script>
    <script src="app.js"></script>
    <style>
        .checkbox {
            margin: 5px 0;
        }
        table {
            width: 100%;
            border-collapse: collapse;
            margin-top: 20px;
        }
        th, td {
            padding: 10px;
            text-align: left;
        }
        th {

```

```

        background-color: #f2f2f2;
    }
</style>
</head>
<body ng-controller="myCtrl">
    <div>
        <label for="hike">Percentage:</label>
        <input type="number" id="hike" ng-model="hike" />
        <button ng-click="hikeSal()">Hike the salary</button>
    </div>
    <div>
        <h3>Hike by dept:</h3>
        <div class="checkbox" ng-repeat="dept in departments track by $index">
            <label>
                <input type="checkbox" ng-model="dept.selected" ng-click="sortEmp(dept.id)" />
                {{dept.name}}
            </label>
        </div>
    </div>
    <table border="1">
        <thead>
            <tr>
                <th>Name</th>
                <th>Dept</th>
                <th>Hike Percentage</th>
                <th>Hike Amount</th>
                <th>Salary</th>
            </tr>
        </thead>
        <tbody>
            <tr ng-repeat="emp in empFiltered track by $index">
                <td>{{emp.name}}</td>
                <td>{{emp.dept}}</td>
                <td>{{emp.hike}}</td>
                <td>{{emp.hikeAmt}}</td>
                <td>{{emp.sal | currency: '₹ '}}</td>
            </tr>
        </tbody>
    </table>
</body>
</html>

```

PROGRAM:

DATABASE QUERY:

```
CREATE DATABASE students;
USE students;
CREATE TABLE studetails (
    id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT
);
INSERT INTO studetails (id, name, age) VALUES
(1, 'James', 25),
(2, 'Smith', 30);
```

Index.html:

```
<html>
  <head>
    <title>TODO supply a title</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
  </head>
  <body>
    <h1>Welcome to the Web Application</h1>
    <p>Click the button below to view data from the database.</p>
    <a href="displayData.jsp">View Data</a>
  </body>
</html>
```

displayData.jsp:

```
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Database Data</title>
  <style>
    table {
      border-collapse: collapse;
      width: 80%;
      margin: 20px auto;
```

```

        font-family: Arial, sans-serif;
    }
    table, th, td {
        border: 1px solid black;
    }
    th, td {
        padding: 10px;
        text-align: center;
    }
    th {
        background-color: #f2f2f2;
    }
</style>
</head>

<body>
    <h1 style="text-align: center;">Data from Database</h1>
    <%
        String jdbcURL = "jdbc:mysql://localhost:3306/student";
        String dbUser = "root";
        String dbPassword = "ifet";
        Connection conn = null;
        Statement stmt = null;
        ResultSet rs = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            conn = DriverManager.getConnection(jdbcURL, dbUser, dbPassword);
            String query = "SELECT * FROM stuDetails"; // Replace with your table name
            stmt = conn.createStatement();
            rs = stmt.executeQuery(query);
            out.println("<table>");
            out.println("<tr>");
            out.println("<th>ID</th>");
            out.println("<th>Name</th>");
            out.println("<th>Age</th>"); // Add more columns as needed
            out.println("</tr>");
            while (rs.next()) {
                out.println("<tr>");
                out.println("<td>" + rs.getInt("id") + "</td>");
                out.println("<td>" + rs.getString("name") + "</td>");
                out.println("<td>" + rs.getInt("age") + "</td>");
                out.println("</tr>");
            }
            out.println("</table>");
        }
    <%>

```



```
    } catch (Exception e) {
        out.println("<p style='color: red;'>Error: " + e.getMessage() + "</p>");
    } finally {
        try {
            if (rs != null) rs.close();
            if (stmt != null) stmt.close();
            if (conn != null) conn.close();
        }
    }
}
```

```
catch (SQLException e) {
    out.println("<p style='color: red;'>Error closing connection: " + e.getMessage() +
"</p>");
}
}
%>
</body>
</html>
```

PROGRAM:

SessionHandlingServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

// Map the servlet to the URL using @WebServlet annotation
@WebServlet("/SessionHandlingServlet")
public class SessionHandlingServlet extends HttpServlet {
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response) throws
IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        // Get username from the form or default to Guest
        String username = request.getParameter("username");
        if (username == null || username.isEmpty()) {
            username = "Guest";
        }

        // Session Handling Methods

        // 1. URL Rewriting
        String urlRewritingLink = "SessionHandlingServlet?username=" + username;

        // 2. Hidden Form Field
        String hiddenForm = "<form action='SessionHandlingServlet' method='post'>"
            + "<input type='hidden' name='username' value='" + username + "'>"
            + "<button type='submit'>Submit with Hidden Field</button></form>";

        // 3. Cookies
        Cookie userCookie = new Cookie("username", username);
        userCookie.setMaxAge(3600); // Expires in 1 hour
        response.addCookie(userCookie);

        // 4. HTTP Session
        HttpSession session = request.getSession();
        session.setAttribute("username", username);

        // Response Output
```

```

        out.println("<html><body>");
        out.println("<h1>Welcome, " + username + "</h1>");
        out.println("<h2>Session Handling Demonstration</h2>");
        out.println("<ul>");
        out.println("<li><a href='" + urlRewritingLink + "'>URL Rewriting</a></li>");
        out.println("<li>" + hiddenForm + "</li>");
        out.println("<li>Cookie Set: " + userCookie.getValue() + "</li>");
        out.println("</ul>");
        out.println("<h3>Session Details:</h3>");
        out.println("<p>Session ID: " + session.getId() + "</p>");
        out.println("<p>Username in session: " + session.getAttribute("username") + "</p>");
        out.println("</body></html>");
    }

```

@Override

```

protected void doGet(HttpServletRequest request, HttpServletResponse response) throws
IOException {
    doPost(request, response);
}
}

```

index.html:

```

<!DOCTYPE html>
<html>
<head>
    <title>Session Handling Example</title>
</head>
<body>
    <h1>Session Handling Demonstration</h1>
    <form action="SessionHandlingServlet" method="post">
        <label for="username">Enter your username:</label>
        <input type="text" id="username" name="username" placeholder="Enter your name">
        <button type="submit">Submit</button>
    </form>
</body>
</html>

```